

Project Report: House Price Predictor - ML / AI Engineer

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Family:	ML / AI Engineer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

End-to-end regression with scikit-learn: EDA notebook, feature engineering, model training, evaluation.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

House Price Predictor** is a beginner-level end-to-end project for ML / AI Engineer. Duration: 2 weeks. End-to-end regression with scikit-learn: EDA notebook, feature engineering, model training, evaluation. Tools: Python, scikit-learn, Jupyter, matplotlib.

Code Reference:

Objective: Exploratory data analysis notebook
Objective: Train/test split with preprocessing pipeline
Objective: Evaluate RMSE and R^2
Objective: Save model with joblib

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

ML / AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a ML / AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: EDA [Beginner]

Chapter focus: Phase: EDA** *(Level: Beginner)

Jupyter exploration. Complete these tasks: Missing values; Correlations. Deliverable for this phase: EDA notebook.

Code Reference:

- [] Missing values
- [] Correlations

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase EDA is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the EDA notebook to the teacher.

Chapter 3: Phase: Features [Beginner]

Chapter focus: Phase: Features** *(Level: Beginner)

Pipeline with ColumnTransformer. Complete these tasks: Scale numerics; Encode categoricals. Deliverable for this phase: Feature pipeline.

Code Reference:

- [] Scale numerics
- [] Encode categoricals

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Features is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Feature pipeline to the teacher.

Chapter 4: Phase: Train [Beginner]**Chapter focus: Phase: Train** *(Level: Beginner)**

RandomForest/LinearRegression. Complete these tasks: Cross-validation; Hyperparams. Deliverable for this phase: Trained model.pkl.

Code Reference:

- [] Cross-validation
- [] Hyperparams

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Train is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Trained model.pkl to the teacher.

Chapter 5: Phase: Evaluate [Beginner]**Chapter focus: Phase: Evaluate** *(Level: Beginner)**

Metrics and residual plot. Complete these tasks: RMSE; Feature importance. Deliverable for this phase: Evaluation report.

Code Reference:

- [] RMSE
- [] Feature importance

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Evaluate is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Evaluation report to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: Jupyter notebook; model.pkl; Evaluation report. Skills demonstrated: scikit-learn, pandas, EDA. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Jupyter notebook
Deliverable: model.pkl
Deliverable: Evaluation report

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Jupyter notebook and passes the rubric.